ECO 310 Problem Set 3 - Problem 1

WithAI Research

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ECO 310 Problem Set 3 - Problem 1 Solution

Cost Functions and Monopoly

Due: April 2, 2026

Part (a): Finding the Cost Function

Given:

- Production technology: $q = 1 + \sqrt{L}$
- Wage rate: $w = \$2$ per labor unit

Find: The firm's cost function $C(q)$

Solution:

To find the cost function, we need to:

1. Invert the production function to express L as a function of q
2. Multiply by the wage rate

Step 1: Invert the production function

Starting with the production function:

$$q = 1 + \sqrt{L}$$

Solving for L:

$$q - 1 = \sqrt{L} \quad (q - 1)^2 = L$$

So the labor demand function is: $\mathbf{L(q) = (q - 1)^2}$

Step 2: Calculate the cost function

Since each labor unit costs \$2:

$$C(q) = w \times L(q) = 2(q - 1)^2$$

Domain note: Since $\sqrt{L} \geq 0$, we have $q \geq 1$. Notice that when $L = 0$, the firm produces $q = 1$ unit (this is like a "free" output from the production technology).

Answer: $C(q) = 2(q - 1)^2$ for $q \geq 1$

Equivalently, expanding: $\mathbf{C(q) = 2q^2 - 4q + 2}$

Part (b): Monopolist's Optimal Quantity and Price

Given:

- The firm is a monopolist
- Demand curve: $Q^D(P) = 200 - 5P$
- Cost function from part (a): $C(q) = 2(q-1)^2$

Find: How much will the monopolist sell, and at what price?

Solution:

Step 1: Find the inverse demand curve

From Q^D(P) = 200 - 5P, solve for P:

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Q = 200 - 5P 5P = 200 - Q P(q) = 40 - q/5
```

Step 2: Write the profit function

Revenue:

```
R(q) = P(q) × q = (40 - q/5) × q = 40q - q²/5
```

Expanding the cost:

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C(q) = 2(q - 1)² = 2q² - 4q + 2
```

Profit:

```
π(q) = R(q) - C(q) = 40q - q²/5 - 2q² + 4q - 2 = 44q - q²/5 - 10q²/5 - 2 = 44q - 11q²/5 - 2
```

Step 3: Maximize profit (first-order condition)

```
dπ/dq = 44 - 22q/5 = 0 44 = 22q/5 220 = 22q q* = 10
```

Step 4: Find the optimal price

```
P* = 40 - q*/5 = 40 - 10/5 = 40 - 2 = 38
```

Verification:

- Second-order condition: $d^2\pi/dq^2 = -22/5 < 0$ ✓ (maximum)
- Demand check: Q^D(38) = 200 - 5(38) = 10 ✓

Answer:

- **Quantity sold: q* = 10 units**
- **Price: P* = $38**

Economic Interpretation

The monopolist restricts output to 10 units (compared to what would occur in perfect competition) and charges $38 per unit. The monopolist earns profit:

```
π* = R(10) - C(10) = 40(10) - 100/5 - 2(10-1)² = 400 - 20 - 162 = $218
```

Files:

- LaTeX source: *src/ps3*problem1*solution.tex
- This markdown: ps3*problem1*solution.md